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WiFi Training Program

Assignment Questions - Module 1

1. In which OSL layer the Wi-Fi standard/protocol fits.

Since Wi-fi is a secure, wireless medium of data transmission, it operates in the Physical and Data-Link Layer of the OSI model.

- ❖ In Wi-fi, the data is transmitted over the air using radio waves and several modulation techniques. So, the data transmission part of Wi-fi falls under the physical layer of the OSI model.
- ❖ Wi-fi does Frame Formatting, Addressing, Error Detection and Correction (CRC) and uses CSMA/CA for collision avoidance, typically highlighting the MAC sublayer functions of the Data-Link Layer of the OSI model.
- ❖ Finally, Wi-fi Does flow control and uses protocols like IP, ARP etc for proper delivery of data which is the LLC sublayer of the data link layer.

2. Can you share the Wi-Fi devices that you are using day to day life, share that device's wireless capability/properties after connecting to the network. Match your device to corresponding Wi-Fi Generations based on properties.

Some of the Devices I use day-to-day in my life include:

A. Smartphone:

- o Standard: **Wi-Fi 6** (IEEE802.11ax)
- o Frequency: **5 GHz**
- o Channel width: 20 / 40 / 80 MHz
- o Link speed: 25 Mbps – 30 Mbps
- o MIMO: **2×2 MIMO**
- o Security: WPA2 / WPA3

B. Laptop:

- o Standard: **Wi-Fi 5** (IEEE802.11ac)

- Frequency: **5 GHz (preferred)** + 2.4 GHz fallback
- Channel width: up to **80 MHz / 160 MHz**
- Link speed: 25 Mbps – 30 Mbps
- MIMO: **2×2 or 3×3**
- Modulation: 256-QAM (Wi-Fi 5), 1024-QAM (Wi-Fi 6)

C. Smart TV:

- Standard: **Wi-Fi 4** (IEEE802.11n)
- Frequency: Mostly **2.4 GHz**
- Channel width: 20 / 40 MHz
- Link speed: 30 Mbps
- MIMO: usually **1×1 or 2×2**

3. What is BSS and ESS?

A Basic Service Set consists of one Access point and Multiple client devices. This Wi-fi network is identified by SSID and BSSID. Whereas, Extended Service Set consists of Multiple Aps connected by a wired lan. Every AP is connected to multiple client devices. All the Aps share a common SSID. ESS enables easy roaming without getting disconnected as while moving, the device automatically gets connected to another AP.

4. What are the basic functionalities of Wi-Fi Access point?

Functionalities of AP:

- Converts wired Ethernet → wireless signals
- Creates the Wi-Fi network (SSID)
- Handles Authentication using Security Protocols like WPA2/WPA3.
- Does frame forwarding and bridging by connecting wireless clients with wired switched/ routers.
- Does Media Access control for avoiding collisions using CSMA/CA protocol.
- Offers roaming support by helping devices to switch between Aps in ESS.
- Controls channel parameters like Bandwidth and Data transmission Rate.

5. What are the differences between Bridge mode and Repeater mode?

Bridge Mode	**Repeater Mode**
Connect two Networks	Extend Network Coverage of a given wi-fi network.
No clients are connected in this mode of operation.	Clients are present.
No major Loss is incured	Loss might occur and re-transmission may be required
Point-to-Point Link	Signal Re-Transmission
Usually used in a building-to-building link.	Used for eliminating dead zones where signal gets completely lost.

6. What are the differences between 802.11a and 802.11b?

IEEE802.11a	**IEEE802.11b**
Operates in the frequency band of 5 Ghz	Operates in the frequency band of 2.4 Ghz
Supports a maximum data rate of 54 mbps	Supports a maximum data rate of 11 mbps
Uses OFDM Modulation Technique	Uses DSSS Modulation technique
Operates in shorter range and offers very less interference	Operates in a longer range but involves more interference
Expensive	Cheap
Not Backward compatible.	Backward compatible with lower versions

7. Configure your modem/hotspot to operate only in 2.4Ghz and connect your laptop/Wi-Fi device, and capture the capability/properties in your Wi-Fi device. Repeat the same in 5Ghz and tabulate all the differences you observed during this

8. What is the difference between IEEE and WFA

IEEE	**WFA**
Institute of Electrical and Electronics Engineers	Wi-Fi Alliance
Global professional/standards organization	Industry Based
Develops technical standards (e.g., 802.11)	Tests and certifies Wi-fi devices.
Theory, protocols, specifications	Practical Implementation based.
Standards like 802.11a/b/g/n/ac/ax	Certifications like Wi-Fi 4, 5, 6
Defines how wifi should work	Ensures devices actually work together

9. List down the type of Wi-Fi internet connectivity backhaul, share your home/college's wireless internet connectivity backhaul name and its properties.

Wi-fi Internet Backhaul Includes the Following.

1. Wired Ethernet Backhaul
2. Fibre Optic Backhaul
3. P2P or multipoint Backhaul
4. Cellular Backhaul
5. DSL Backhaul

In my house, Fibre Optic Backhaul is used and its properties are:

- Medium: Optical fiber
- Speed: 100 Mbps – 1 Gbps (or more)
- Latency: Very low (≈5–20 ms)
- Reliability: High
- Device chain: ISP Fiber → ONT/Modem → Wi-Fi Router → Devices

10. List down the Wi-Fi topologies and use cases of each one.

- Infrastructure Mode - Home/Office Wi-fi Routers
- Mobile-Hotspot Mode - Sharing Mobile Data between two smart phones
- Repeater Mode - Eliminating Dead Zones
- Bridge Mode - Linking two buildings in a campus without laying cables
- Mesh Mode - Large homes or offices needing uninterrupted roaming
- Work-Group Mode - Small office sharing a printer and files over Wi-Fi
- Ad-Hoc Mode - Quick file transfer between two laptops without a router
- IOT Gateway Mode - Smart home hub connecting sensors (lights, thermostat) to the cloud